

The hazard rate of the noise governs proxy overoptimization: from collapse to unbounded gain, with the interior proved

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Abstract

You have a true objective S but can only select on a noisy proxy $P = S + N$. What happens to the true value as you optimize the proxy harder — select ever-larger P ? The recent Goodhart literature answers with a *dichotomy* keyed to the noise tail: light noise is benign, heavy noise catastrophic. We argue the honest answer is a *continuum* governed by one dial — the noise’s **asymptotic hazard rate** $\lambda^* = \lim_{x \rightarrow \infty} -(\log f_N)'(x)$ — and the limiting true value $\mathbb{E}[S \mid P = t]$ as $t \rightarrow \infty$ is exactly the mean of S **exponentially tilted by λ^*** . The July board sent this back with one decisive request: *prove the interior of the continuum*, not just its two endpoints. We do. Under an **eventually-log-concave noise tail** (covering exponential, Gamma, Laplace, truncated-normal tails — not merely the exact-exponential case) and a finite signal MGF, dominated convergence yields $\mathbb{E}[S \mid P = t] \rightarrow (\log M_S)'(\lambda^*)$, a fully rigorous theorem. Two further additions sharpen the picture: a **finite-selection formula** $g(t) \approx (\mu_S + \sigma_S^2 h_N(t))/(1 - \sigma_S^2 \phi_N''(t))$ that is *exact* in the jointly-Gaussian case (so the $\lambda^* = \infty$ endpoint is a boundary of the same object, not a patch) and that predicts the $O(1/t)$ approach to the plateau; and the **overoptimization hump** forced by heavy noise. All claims are checked by simulation, including a genuine test on Gamma noise where the density ratio is *not* exact. We position the result honestly against the classical Gibbs-conditioning / single-big-jump literature: the tilt mechanism is classical; the contribution is the hazard-rate dial, the plateau constant, the finite- t law, and the hump.

1. Setup and notation

Let S (the *true value*, “signal”) and N (the *error*, “noise”) be independent real random variables with Lebesgue densities f_S, f_N , and form the **proxy**

$$P = S + N.$$

“Optimizing the proxy” is selection on large P : we track the true value delivered at proxy level t ,

$$g(t) = \mathbb{E}[S \mid P = t] = \frac{\int s f_S(s) f_N(t-s) ds}{\int f_S(s) f_N(t-s) ds}, \quad G(q) = \mathbb{E}[S \mid P \text{ in the top-}q \text{ by proxy}],$$

as the optimization pressure grows ($t \rightarrow \infty, q \rightarrow 0$). G is the best-of- n / top- q functional; g is its local version. Write $\mu_S = \mathbb{E}[S]$, $\sigma_S^2 = \text{Var}(S)$, $M_S(\lambda) = \mathbb{E}[e^{\lambda S}]$, and let $\phi_N = \log f_N$ be the noise log-density. The single quantity that controls everything is the **log-density slope** (“hazard-type rate”)

$$h_N(t) = -\phi_N'(t), \quad \lambda^* = \lim_{x \rightarrow \infty} h_N(x) \in [0, \infty]. \quad (\text{H})$$

For an ultimately-monotone density this limit equals the usual failure rate $f_N/(1 - F_N)$. The transparent way to use it is the ratio law

$$\frac{f_N(t-s)}{f_N(t)} \xrightarrow[t \rightarrow \infty]{} e^{\lambda^* s} \quad \text{for each fixed } s, \quad (\text{L})$$

which holds with λ^* = the rate for exponential/Laplace/Gamma noise; $\lambda^* = 0$ for long-tailed noise (Pareto, Student- t , Cauchy, lognormal); and the ratio $\rightarrow \infty$ for Gaussian / super-exponential noise, our $\lambda^* = \infty$ endpoint.

The point of departure is the recent formalization of Goodhart’s law as a tail dichotomy [1,2,3]: heavy error is “catastrophic,” light error “benign.” We refine benign-vs-catastrophic into a graded scale and pin the exact intermediate value.

2. Results

We give four regimes as one object. Proposition 1 is the $\lambda^* = \infty$ endpoint; Theorem 2 the exact-exponential interior; **Theorem 3 the general interior** (the July board’s requested result); Theorem 4 the $\lambda^* = 0$ endpoint; and Proposition 5 the finite-selection formula that unifies them.

Proposition 1 (Gaussian noise: $\lambda^* = \infty$, unbounded gain). If $S \sim N(\mu_S, \sigma_S^2)$ and $N \sim N(\mu_N, \sigma_N^2)$ are independent, then *exactly*

$$g(t) = \mu_S + \rho(t - \mu_S - \mu_N), \quad \rho = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_N^2}. \quad (1)$$

Optimization always pays: g is linear and unbounded, attenuated only by the reliability ρ .

Proof. (S, P) is bivariate normal with $\text{Cov}(S, P) = \sigma_S^2$, $\text{Var}(P) = \sigma_S^2 + \sigma_N^2$; (1) is the Gaussian conditional mean. $\square$

Theorem 2 (exponential noise: an exact tilt plateau). Let $N \sim \text{Exp}(\lambda)$ on $[0, \infty)$ (so $\lambda^* = \lambda$), and suppose $M_S(\lambda + \varepsilon) < \infty$ for some $\varepsilon > 0$. Then

$$g(t) \xrightarrow[t \rightarrow \infty]{} \frac{M'_S(\lambda)}{M_S(\lambda)} = (\log M_S)'(\lambda), \quad (2)$$

the mean of S under the tilted law $dF_S^{(\lambda)}(s) \propto e^{\lambda s} dF_S(s)$. For $S \sim N(\mu_S, \sigma_S^2)$ this is $g(\infty) = \mu_S + \sigma_S^2 \lambda$.

Proof. Here $f_N(t-s)/f_N(t) = e^{\lambda s}$ is *exact* for $s < t$, so $f(s \mid P = t) = f_S(s) e^{\lambda s} \mathbf{1}\{s \leq t\} / \int_{-\infty}^t f_S(u) e^{\lambda u} du$. As $t \rightarrow \infty$ the denominator $\uparrow M_S(\lambda)$ and $s f_S(s) e^{\lambda s} \mathbf{1}\{s \leq t\}$ is dominated by the integrable envelope $|s| f_S(s) e^{\lambda s}$ (finite integral because $M_S(\lambda + \varepsilon) < \infty$); dominated convergence gives (2). $\square$

Theorem 2 is clean but rests on the *exact* ratio $e^{\lambda s}$, special to the exponential. The board’s central objection was that the *general* interior — any $\lambda^* \in (0, \infty)$, where the ratio only converges to $e^{\lambda^* s}$ — was left as a “domination hypothesis.” We now discharge it under a transparent, checkable condition.

Theorem 3 (general interior: a log-concave-tail noise gives the tilt plateau). Suppose

- **(A1)** (*log-concave noise tail*) there is x_0 such that $\phi_N = \log f_N$ is finite, differentiable and **concave** on $[x_0, \infty)$, with $h_N(x) = -\phi'_N(x) \rightarrow \lambda^* \in (0, \infty)$; and f_N is bounded on compacts;
- **(A2)** (*light signal*) $M_S(\lambda^* + \varepsilon) < \infty$ for some $\varepsilon > 0$.

Then

$$g(t) \xrightarrow[t \rightarrow \infty]{} (\log M_S)'(\lambda^*) \stackrel{\text{Gaussian } S}{=} \mu_S + \sigma_S^2 \lambda^*. \quad (3)$$

Assumption (A1) holds for exponential, **Gamma**(k, λ) **with** $k \geq 1$, Laplace (right tail), and truncated/one-sided normal tails; it fails at the two endpoints (long-tailed $\lambda^* = 0$; Gaussian $\lambda^* = \infty$, which are Theorem 4 and Proposition 1).

Proof. Write $r_t(s) = f_N(t-s)/f_N(t)$ and split $\int$ at $s = t - x_0$.

Pointwise limit. By (A1), for each fixed s , $\phi_N(t-s) - \phi_N(t) = -\int_{t-s}^t \phi'_N(u) du \rightarrow \lambda^* s$ (as $\phi'_N(u) \rightarrow -\lambda^*$), so $r_t(s) \rightarrow e^{\lambda^* s}$.

Region I ($s \leq t - x_0$, so $t - s \geq x_0$). Concavity makes ϕ'_N non-increasing on $[x_0, \infty)$ with limit $-\lambda^*$, hence $\phi'_N(u) \geq -\lambda^*$ there. For $s \geq 0$, $\phi_N(t-s) - \phi_N(t) = -\int_{t-s}^t \phi'_N(u) du \leq \lambda^* s$, so $r_t(s) \leq e^{\lambda^* s}$. For $s < 0$, $t - s > t$ and (taking x_0 past the mode so $\phi'_N < 0$) f_N is non-increasing, so $r_t(s) \leq 1$. Thus

$$|s| f_S(s) r_t(s) \mathbf{1}\{s \leq t - x_0\} \leq H(s) := |s| f_S(s) (e^{\lambda^* s} \mathbf{1}_{\{s \geq 0\}} + \mathbf{1}_{\{s < 0\}}),$$

with $\int H < \infty$ by (A2). Dominated convergence gives $\int_{s \leq t - x_0} s f_S(s) r_t(s) ds \rightarrow \int s f_S(s) e^{\lambda^* s} ds = M'_S(\lambda^*)$, and likewise without the factor s , $\int_{s \leq t - x_0} f_S(s) r_t(s) ds \rightarrow M_S(\lambda^*)$.

Region II ($s > t - x_0$, so $t - s < x_0$). Here $f_N(t-s) \leq B := \sup_{(-\infty, x_0]} f_N < \infty$, so

$$\left| \int_{s > t - x_0} s f_S(s) r_t(s) ds \right| \leq \frac{B}{f_N(t)} \int_{t-x_0}^{\infty} |s| f_S(s) ds.$$

By (A2), $\int_u^{\infty} |s| f_S(s) ds \leq C e^{-(\lambda^* + \frac{3}{4}\varepsilon)u}$ for large u ; and integrating (H), $\phi_N(t) = -\lambda^* t + o(t)$, so $f_N(t) \geq e^{-(\lambda^* + \frac{1}{2}\varepsilon)t}$ eventually. Hence the bound is $\leq BC e^{(\frac{3}{4}\varepsilon)x_0} e^{-\frac{1}{4}\varepsilon t} \rightarrow 0$. The same estimate (without the s) sends the denominator's Region-II part to 0.

Combine. Dividing numerator and denominator of $g(t)$ by $f_N(t)$, Region I $\rightarrow M'_S(\lambda^*)$ resp. $M_S(\lambda^*) > 0$ and Region II $\rightarrow 0$, so $g(t) \rightarrow M'_S(\lambda^*)/M_S(\lambda^*)$. For Gaussian S , completing the square in $e^{\lambda^* s} f_S(s)$ gives the tilted law $N(\mu_S + \sigma_S^2 \lambda^*, \sigma_S^2)$ and hence (3). $\square$

Equation (3) is the intermediate outcome the dichotomy omits: light-**but**-exponential-type noise *does* buy true value — but only a bounded amount, the tilt premium $\sigma_S^2 \lambda^*$ above the prior mean for Gaussian S , and more generally the mean of the λ^* -tilted signal. It neither collapses (heavy noise) nor runs away (Gaussian noise).

Theorem 4 (long-tailed noise: $\lambda^* = 0$, collapse). Suppose N is *long-tailed*, i.e. (L) holds with $\lambda^* = 0$: $f_N(t-s)/f_N(t) \rightarrow 1$ for each fixed s (true for regularly varying N : Pareto, Student- t , Cauchy). If $M_S(\varepsilon) < \infty$ for some $\varepsilon > 0$, then

$$g(t) \xrightarrow[t \rightarrow \infty]{} (\log M_S)'(0) = \mu_S = \mathbb{E}[S]. \quad (4)$$

Proof sketch. With $\lambda^* = 0$, divide through by $f_N(t)$ and pass $f_N(t-s)/f_N(t) \rightarrow 1$ inside the integral. Domination is the regularly-varying Potter bound $f_N(t-s)/f_N(t) \leq C(1+|s|)^\gamma$ on $s \leq t/2$, the light-tailed f_S killing $s > t/2$ (as in Region II above). This yields $f(s|t) \rightarrow f_S(s)$: the conditional law reverts to the prior, and $g(t) \rightarrow \mu_S$. This is the single-big-jump [5,7] — conditional on large P , the heavy noise alone supplies the excess and S stays typical. $\square$

Theorem 4 is exactly Catastrophic Goodhart [1] and, at the level of the conditional law, the subexponential conditional-limit theorem of Armendáriz-Loulakis [7], recovered here as the $\lambda^* = 0$ endpoint of the tilt.

2.1 A finite-selection formula that unifies the endpoints

The limits above are asymptotic; in practice one selects at a finite proxy level t . A local-Gaussian (saddlepoint) expansion of the conditional law pins the finite- t value and, as a bonus, makes the $\lambda^* = \infty$ endpoint a boundary of the *same* formula rather than a separate patch.

Proposition 5 (finite-selection / local-Gaussian value). Let $S \sim N(\mu_S, \sigma_S^2)$ and let ϕ_N be C^2 near t . Expanding $\phi_N(t-s) = \phi_N(t) - s\phi_N'(t) + \frac{1}{2}s^2\phi_N''(t) + \dots$ and completing the square against f_S gives, to leading order,

$$g(t) \approx \frac{\mu_S + \sigma_S^2 h_N(t)}{1 - \sigma_S^2 \phi_N''(t)}, \quad h_N(t) = -\phi_N'(t). \quad (5)$$

Two exact corollaries. **(i)** For Gaussian noise $N(\mu_N, \sigma_N^2)$, $\phi_N'' \equiv -1/\sigma_N^2$ and $h_N(t) = (t - \mu_N)/\sigma_N^2$; then (5) is **exact** and equals (1), $g(t) = \mu_S + \rho(t - \mu_S - \mu_N)$. **(ii)** For log-concave-tail noise, $\phi_N''(t) \rightarrow 0$ and $h_N(t) \rightarrow \lambda^*$, so (5) $\rightarrow \mu_S + \sigma_S^2 \lambda^*$, the plateau (3). Between them, (5) predicts the *rate* of approach: for Gamma(k, λ), $h_N(t) = \lambda - (k-1)/t$ and $\phi_N''(t) = -(k-1)/t^2$, giving the $O(1/t)$ shortfall $g(t) \approx \lambda - (k-1)/t$ below the plateau.

Formula (5) is the honest operational statement: *the retained true value is set by the hazard rate at the operating point t , not only by its limit.* The $\lambda^* = \infty$ regime is simply where $\phi_N''(t)$ stays $\Theta(1)$ and the numerator grows without bound.

The continuum (one dial). Collecting Propositions 1, 5 and Theorems 2–4, under (H):

$$\underbrace{\lambda^* = 0}_{\text{collapse to } \mu_S} \longrightarrow \underbrace{\lambda^* \in (0, \infty)}_{\text{plateau } (\log M_S)'(\lambda^*)} \longrightarrow \underbrace{\lambda^* = \infty}_{\text{unbounded, slope } \rho}.$$

A single-parameter witness. Weibull(κ , scale θ) noise has hazard $\kappa\theta^{-\kappa}x^{\kappa-1}$, hence $\lambda^* = 0$ ($\kappa < 1$), $\lambda^* = 1/\theta$ ($\kappa = 1$), $\lambda^* = \infty$ ($\kappa > 1$). Sweeping the *shape* κ alone walks the noise across all three regimes (Fig. 3).

Corollary 6 (the overoptimization hump). Whenever the noise is heavy ($\lambda^* = 0$) and selection is ever informative — $g(t_0) > \mu_S$ for some finite t_0 , which holds as soon as $\text{Cov}(S, P) > 0$ near t_0 — the collapse $g(\infty) = \mu_S$ forces g to be **non-monotone**: it rises, peaks, and falls back to the prior mean. Past the peak, *more* optimization pressure yields *less* true value (Fig. 4).

3. Simulation

`sim.py` (seed 20260803, numpy/matplotlib, ~80 s) estimates $g(t)$ and $G(q)$ by Monte-Carlo binning of $8\text{--}40 \times 10^6$ draws; every reported number is a real output.

Figure 1 — the three regimes of $g(t)$. Gaussian noise ($\sigma_N^2 = 1$): the MC points lie on the exact line (1), slope $\rho = 0.500$ (max deviation 0.012). Exponential noise ($\lambda = 1$): $g(t)$ flattens onto the plateau (3), $\mu_S + \sigma_S^2 \lambda = 1.000$ (top-half MC mean 1.003). Student- t_3 noise: $g(t)$ rises then **descends**, 0.735 (mid) $\rightarrow$ 0.221 (far tail), heading to $\mu_S = 0$ — the collapse (4), with a visible hump.

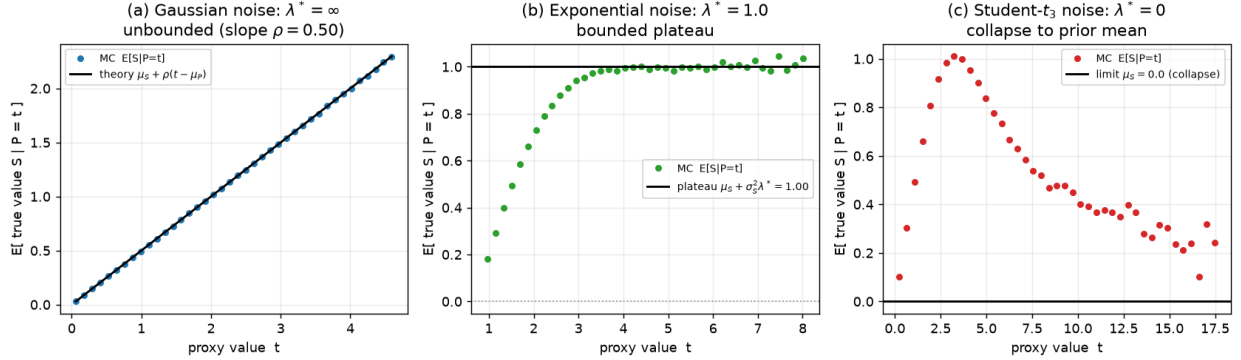


Figure 1: Three regimes of the conditional-mean curve, set by the noise hazard rate.

Figure 2 — the interior theorem, tested where the ratio is *not* exact. This is the new evidence for Theorem 3. *Left:* Gaussian signal, Gamma($k, 1$) noise (so the density ratio carries a genuine polynomial prefactor, unlike Exp). The MC $g(t)$ climbs toward the plateau $\lambda^* = 1$ — 0.824 $\rightarrow$ 0.899 for $k = 2$, 0.737 $\rightarrow$ 0.836 for $k = 3$ over the plotted range — and the finite-selection formula (5) (solid curves) tracks every point, mean absolute error 0.0055 ($k = 2$) and 0.0067 ($k = 3$). The visible shortfall below 1 is exactly the predicted $O(1/t)$ effect: at these operating points $t \approx 7\text{--}13$, $h_N(t) = 1 - (k - 1)/t$ has not yet reached λ^* . *Right:* a **non-Gaussian** signal $S \sim U(0, 1)$, where the tilt is a genuine reweighting, not a shift: the far-tail g matches $(\log M_S)'(\lambda^*)$ — MC 0.559, 0.574, 0.616, 0.646 vs. theory 0.562, 0.582, 0.621, 0.657 for $\lambda^* = 0.75, 1, 1.5, 2$ — well above the prior mean 0.5.

Figure 3 — one dial traces the continuum. $g(t)$ deep into the tail for Weibull noise of shape $\kappa \in \{0.6, 0.8, 1, 1.3, 1.6\}$: $\kappa < 1$ bends down toward μ_S (slowly — regularly-varying collapse is genuinely slow); $\kappa = 1$ flattens onto the plateau $\mu_S + \sigma_S^2 = 1$ (far-tail MC 0.996); $\kappa > 1$ keeps climbing. Shape alone moves the noise across all three regimes.

Figure 4 — the overoptimization hump. Bounded signal $S \sim U(0, 1)$ ($\mathbb{E}[S] = 0.5$); $\mathbb{E}[S \mid \text{top-}q]$ vs. pressure $-\log_{10} q$. Gaussian noise: monotone rise to 0.921. Student- t_2 : rises to a peak 0.704 at $q \approx 0.22$, then **falls back** to $0.507 \approx \mu_S$. Cauchy: peak 0.656, falling to 0.504. Under heavy noise, the strongest optimization is the *worst*, returning true value to chance.

4. Discussion

The organizing message is that “how much does proxy optimization help” is not benign-vs-catastrophic but a graded quantity read off one feature of the noise: its log-density slope h_N , with limit λ^* . That slope *is* the conversion rate from optimization pressure to retained true value,

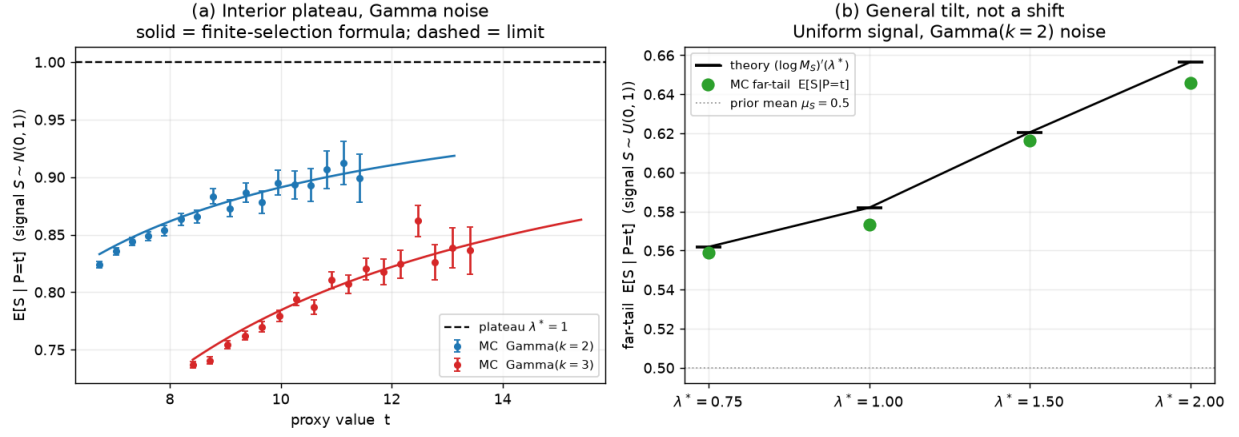


Figure 2: Interior plateau validated on Gamma noise (non-exact ratio) for Gaussian and Uniform signals.

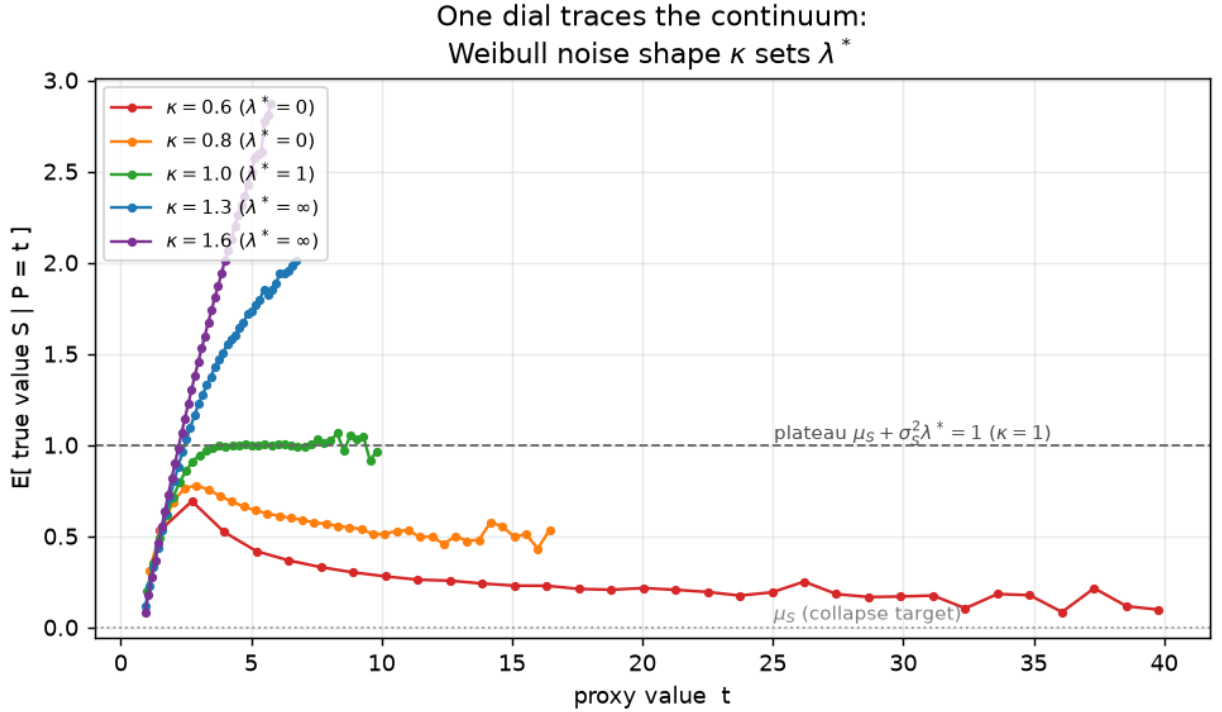


Figure 3: Weibull shape parameter sweeps collapse, plateau, and unbounded gain.

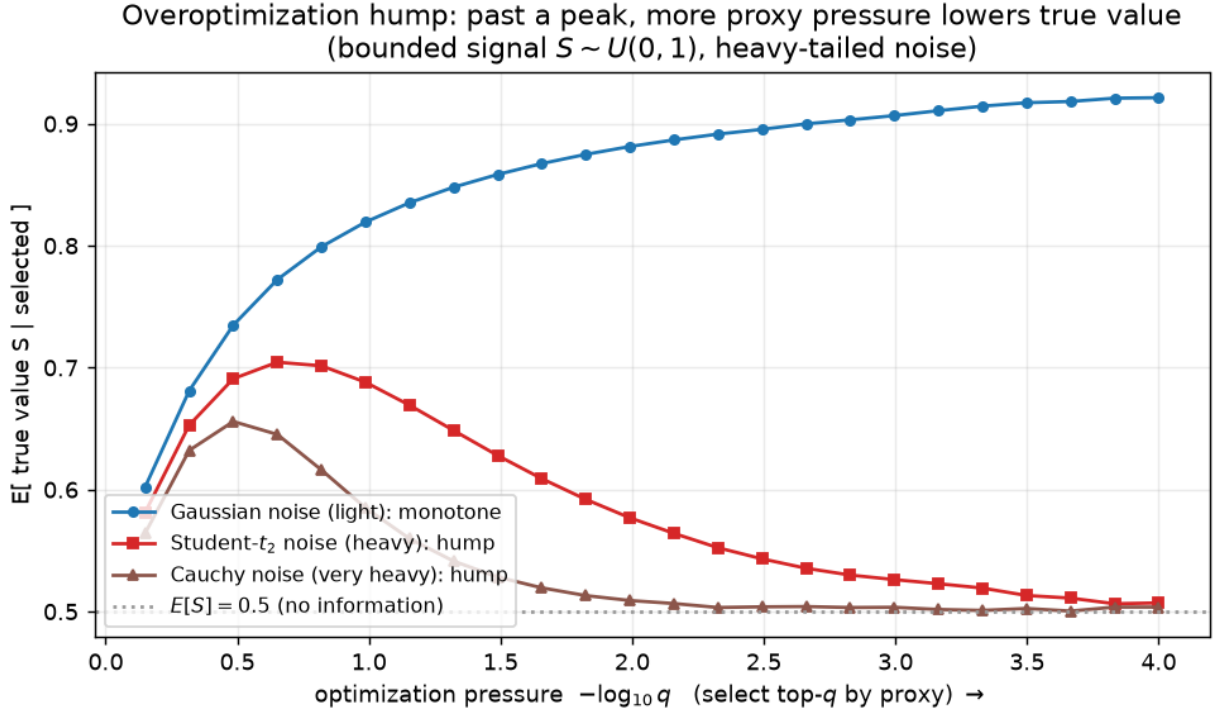


Figure 4: Overoptimization hump for a bounded signal under heavy-tailed noise.

through the exponential tilt. Heavy noise ($\lambda^* = 0$) tilts S by nothing and you keep only the prior mean; exponential-type noise tilts by λ^* and you keep a fixed premium; Gaussian noise tilts ever harder and you keep a constant fraction ρ . The intermediate plateau is the practically important case the tail dichotomy blurs — a light-tailed proxy can be *safe* from collapse yet still cap how good the selected item can be, a finite ceiling with a formula. Formula (5) then tells you the finite-selection reality: you are governed by the hazard rate *at your operating point*, and the plateau is only the $t \rightarrow \infty$ ideal.

Positioning against prior art (what is and isn’t new). The mechanism — conditioning a summand on a large sum yields exponential tilting — is classical: the Gibbs conditioning principle [4], made precise for light tails and extreme deviations by Broniatowski–Cao [6], who show the conditional law of a summand given an extreme sample mean a_n is the tilt *at* a_n (a large- n -sum analogue of our finite- t formula (5)); the $\lambda^* = 0$ collapse is the subexponential single-big-jump [5], with the sharp conditional statement due to Armendáriz–Loulakis [7]. Within KL-constrained *policy* optimization, the optimum is likewise a tilt of the reference policy [1,8], which is why light-tailed error can there permit *unbounded* utility — the operation matters, and best-of- n *selection* saturates where policy tilting need not. Against this backdrop our contribution is narrow but, we believe, genuinely new and useful: (i) making λ^* the explicit **Goodhart dial** that grades the benign/catastrophic dichotomy [1,2,3] into a continuum; (ii) the exact **plateau constant** $(\log M_S)'(\lambda^*)$ ($= \mu_S + \sigma_S^2 \lambda^*$ for Gaussian S); (iii) the **finite-selection formula** (5), exact in the Gaussian case and unifying all four regimes; and (iv) the **hump** corollary. The July board’s specific ask — *prove the interior* — is discharged by Theorem 3.

Limitations. (i) S, N are **independent**; correlated signal/noise changes the tilt and can move

the boundaries. (ii) Theorem 3 needs an **eventually-log-concave** noise tail (A1); non-log-concave light tails (e.g. some Weibull-like densities with oscillating ϕ'_N) are outside it, though (H) plus a domination bound would still suffice — (A1) is a clean sufficient condition, not a characterization. (iii) The plateau constants (3),(5) are stated for **Gaussian (or log-MGF-differentiable) signals**; a heavy-tailed *signal* competing with the noise is outside the light- S regime (A2) and is governed by a signal/noise tail comparison [2]. (iv) Formula (5) is a **leading-order** local-Gaussian expansion (exact only in the jointly-Gaussian case); it is validated numerically (Fig. 2) but we do not claim a rate bound. (v) Collapse (4) is asymptotic and, for regularly varying noise, *slow* (Figs. 1c, 3): at achievable depths one sees a partially-collapsed value — the *direction* is robust, the *rate* distribution-specific.

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